# Sampled or Stated? Post-Training Moves a Language Model's Differential Diagnosis from Its Samples to Its List

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**Abstract**

A language model's differential diagnosis can be read two ways: from the ranked list it states in one answer, or from the spread of its answers over repeated samples. Pass@$k$, self-consistency and sampling-based uncertainty all use the second. Recent work shows that instruction-tuned models cannot sample from distributions they can describe. We test what this means for diagnosis, where a reference differential exists, with base/post-trained pairs from two open families at 7–32B (Qwen2.5-7B and -32B, Qwen3-8B, and every released stage of OLMo 3 7B and OLMo 3.1 32B) and three medical models fine-tuned from one parent, using 32 samples per case on DDXPlus, whose patients come with a reference differential, and on rare-disease cases from Phenopacket Store. The sampled differential collapses: post-trained models trail their base models by 4–28 points at pass@32, give 1.3–3.6 distinct top answers per 32 samples against 3.8–8.2, and gain little from majority voting, and agreement between samples becomes a weaker error signal. The stated list does not collapse. In both families it moves closer to the reference differential and names fewer implausible conditions, robustly to list format, top-1 accuracy and the elicitation prompt; and drawing top answers from a model's own stated lists recovers more coverage than its base model's samples (Qwen3-8B: pass@32 0.81 against 0.73). Whether the stated probability is the better error signal depends on the model; a held-out audit on 100 labeled cases picks the better signal in 82% of draws for post-trained models, against 54% for always using agreement. Which post-training stage narrows the samples most depends on the recipe: the RL stage in OLMo 3 7B, SFT in OLMo 3.1 32B. On real rare-disease cases the same shifts appear without a crossover: voting stops helping and agreement weakens until it ties the stated probability. Neither readout tracks how ambiguous a case is, and on rare disease medical RL raises stated confidence from 0.51 to 0.67–0.69 at about 2% accuracy. Claims about a model's diagnostic breadth should say which readout they use.

## 1 Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. Language-model evaluations in medicine nonetheless score a single answer: multiple-choice accuracy, top-1 diagnosis, or at most pass@$k$ for a small $k$. Post-training, whether general instruction tuning and RLHF or domain-specific medical reinforcement learning and distillation, is reported to improve these scores [1, 9, 17].

A model's differential can be read in two ways: from the ranked list it states in one answer, or from the distribution of its answers over repeated samples. Much current practice uses the second. Pass@$k$, self-consistency voting [30] and sampling-based uncertainty such as semantic entropy [8] all draw on repeated samples, and so does the evidence that reinforcement learning sharpens models,

which win at pass@1 while their base models solve more problems given many samples [23, 32, 34]. The two readouts agree for a model that samples from the distribution it states. Work outside medicine suggests that aligned models do not. They can verbalize distributions they fail to sample from [21, 35], their verbalized confidence is better calibrated than their token probabilities [29], and their samples can homogenize enough to defeat sampling-based uncertainty [18]. Jang et al. [10] show this directly for survey-response distributions: instruction-tuned models sample worse than their own base models, more so at each post-training stage, while still describing the target accurately in a single call. Whether the same holds for a differential diagnosis, where the reference is a clinical distribution with a correct answer rather than an opinion profile, and what it does to the tools clinicians and evaluators build on repeated samples, has not been tested.

We measure it directly. We test it with a design that holds the pretrained model fixed and varies only the post-training: (i) two base/instruct pairs, Qwen2.5-7B and Qwen3-8B, which isolate general post-training; (ii) three public medical models fine-tuned from the same parent, Qwen2.5-7B-Instruct: AlphaMed (rule-based RL), HuatuoGPT-o1 (SFT then PPO), and m1 (SFT distillation), plus AlphaMed's companion checkpoint trained from the base model; (iii) every released stage of OLMo 3 7B [28], from one base model through SFT, DPO and RL, plus a checkpoint trained with RL directly from the base; (iv) at 32B, Qwen2.5-32B with its Instruct model and every released stage of OLMo 3.1 32B. We draw 32 samples per case. On DDXPlus [7], which ships a reference differential for every synthetic patient, we score both readouts against that reference; on rare-disease cases from Phenopacket Store [5] we score accuracy and stated confidence.

**Findings.**

1. **Post-training collapses the sampled differential.** In both Qwen families at 7–8B the post-trained model is far better on its first answer (+13 to +27 points) but its pass@$k$ curve crosses the base model's at $k$≈4–8, and at $k$=32 the base model leads by 10–15 points; at 32B the lead is smaller (4.0–4.9 points). Post-trained models give 1.3–2.6 distinct top answers per 32 samples, against 3.8–8.3, and can't-miss conditions drop out of their samples (§4.1, §4.2, §4.5). Which stage narrows the samples most depends on the recipe: the RL stage applied after SFT and DPO in OLMo 3 7B (19 points of pass@32), SFT in OLMo 3.1 32B (12 points), where the RL stage costs only 3 (§4.4).

2. **The stated differential does not collapse; it improves.** A single stated list from the post-trained Qwen models at 7–32B is closer to the reference differential than the base model's (total-variation distance −0.04 to −0.09), covers 6–11 points more of its mass, and names can't-miss conditions 9–13 points more often while naming fewer implausible ones, and their samples move away from the reference. The same holds for OLMo 3.1 32B. The gain survives restricting to complete lists, stratifying by top-1 correctness and correcting for each readout's floor, and drawing top answers from a model's own stated lists recovers more coverage than its base model's samples (§4.9). OLMo 3 7B's five-item lists do not improve, but its stated distribution does when asked for a probability on every condition (§4.6, §4.8).

3. **Sampling-based methods lose their value; which signal to trust is model-dependent but cheap to check.** Majority voting over 32 samples gains base models 3–8 points at $T$=0.7 and the general post-trained models at most 1.9; models that reason before answering still gain 4–5. For post-trained Qwen models, the top three of one stated list beats the three most frequent of 32 sampled answers by 6–14 points. Agreement between samples becomes a weaker error signal in every family; the stated probability overtakes it for Qwen3-8B and OLMo-3-7B-Instruct but not for the other three general post-trained models. Across 56 model–setting

combinations the winner tracks how often a model's samples all agree and how informative its stated probability is, and a held-out audit on 100 labeled cases picks it in 82% of draws for post-trained models, while always using agreement is right for 54% of them. Neither readout tracks how ambiguous a case truly is (§4.7, §4.10).

4. **On real rare-disease cases the shifts replicate without a crossover.** Voting gains fall from 3.1–4.6 points for base models to 0.4–1.5 after post-training, and agreement weakens as an error signal until it ties the stated probability; 32 samples still beat one list (§4.13).

5. **Medical post-training adds accuracy, not breadth, and raises overconfidence on rare disease.** AlphaMed improves on its parent at every $k$ without widening its sampled differential; its companion checkpoint trained from the base model keeps most of the base model's diversity (§4.11). With accuracy near 2%, medical RL raises stated confidence from 0.51 to 0.67–0.71, and every OLMo stage states 0.73–0.84 at about 1% accuracy (§4.12).

6. **RL from the base model need not collapse the samples.** GRPO from Qwen3-8B-Base nearly matches the official post-trained model's top-1 accuracy while keeping more than twice its sampled diversity, and a group-level reward narrows the samples less than standard RLVR at equal accuracy (§5).

## 2 Related work

**What RL does to a model's output distribution.** Yue et al. [34] find that RL with verifiable rewards improves pass@1 on math and code while base models win at large $k$. Wu et al. [32] argue that RLVR mostly reweights solutions already in the base model's support. Oh et al. [23] document a "sharpening tax" across 14 base/post-trained pairs on agentic benchmarks. In contrast, Cheng et al. [3] report that RL expands the boundary on domains under-represented in pretraining. In medicine, the only pass@$k$ analysis we know of studies vision-language models and finds that RL mainly sharpens [11]. All of these measure the sampled distribution; we also measure the stated one.

**Diversity loss from alignment.** Instruction tuning and RLHF reduce output diversity [14, 22, 31], a loss that grows with scale for SFT [27]. Related work explains this as typicality bias in preference data [35] and documents cross-model homogeneity [12]. Zhang et al. [35] also track diversity across the SFT, DPO and RLVR checkpoints of Tülu 3 on creative writing; we do the same for OLMo 3 against a clinical reference distribution.

**Stated versus sampled distributions.** Meister et al. [21] find that language models asked to verbalize a group's opinion distribution match it better than their samples or log-probabilities do, and that they "struggle to sample from their own distribution"; they study instruct and API models on opinion surveys, which have no correct answer. Verbalized sampling [35] prompts aligned models to state a distribution over candidate responses and recovers part of the diversity lost in post-training. Tian et al. [29] show that RLHF models' verbalized confidence is better calibrated than their token probabilities. Liu [18] report that aligned models' samples often fall into a single semantic cluster, leaving sampling-based uncertainty [8] without signal on those questions; in a Zephyr base/SFT/DPO ablation the collapse appears at DPO. Closest to our question, Jang et al. [10] show on synthetic and survey targets that instruction-tuned models sample worse than their own base models in every family they compare, that the gap grows at each OLMo-2 post-training

stage, and that the same models describe the target accurately in one call. Verbalized-confidence studies reach mixed conclusions: sample consistency usually discriminates better than verbalized confidence [33], including for medical diagnosis [26]. We extend this line to diagnosis, where the reference distribution has a correct answer, test it on base/post-trained pairs at 7–32B and on every stage of two open pipelines, and measure what it does to diagnostic aggregation and uncertainty.

**Medical reasoning models and calibration.** Medical post-training recipes include verifier-guided SFT and PPO [1], minimalist rule-based RL [17], and test-time-scaling distillation [9]. Reward models and RL tend to favor confident answers [4, 15], and reasoning models are often overconfident [20]. LLMs have been evaluated as differential-diagnosis aids on clinicopathological cases [13, 19]. We show that this overconfidence appears in public medical descendants of one parent, on rare disease.

**Differential diagnosis benchmarks and distributional RL.** DDXPlus [7] was introduced for automatic diagnosis systems trained on differentials [6]. Recent LLM evaluations on it score hierarchical top-$k$ accuracy [16] or interactive diagnosis [25]; the dataset's own metrics compare predicted and reference differentials as sets. Puri et al. [24] train models with GRPO to output several answers per forward pass, rewarding coverage of a gold *set* and set-level calibration, and evaluate on DDXPlus. We instead compare stated and sampled differentials against the reference *distribution* across public post-trained model families. Our group-level reward operates on the distribution of independently sampled answers (§5).

## 3 Experimental setup

**Tasks.** **DDXPlus** [7]: synthetic patients produced by a rule-based medical knowledge base. Each patient has a true pathology and a reference differential, a probability distribution over the 49 conditions. We use 980 test patients, 20 per condition. Each patient's coded symptoms and history are rendered as a short vignette; we fix known mistranslations in the English evidence descriptions. The prompt lists all 49 conditions and asks for the five most likely, with probabilities, as JSON. **Phenopacket Store** [5] (v0.1.27): published rare-disease cases annotated with HPO phenotypes and an OMIM diagnosis. We use a disease-balanced set of 1,401 cases covering 749 diseases (at most two per disease). Each case is a vignette of present and absent phenotypes, and the model is asked for an open-ended differential. Answers are graded by mapping names to Mondo; we credit an exact match, and separately a "related" match to a specific ancestor group or subtype.

**Models.** Table 1. All checkpoints are public. For the medical models we verified the parent model from each released configuration.

**Sampling.** We draw $n$=32 samples per case with vLLM, at $T$=0.7, top-$p$ 0.95, and at $T$=1.0, top-$p$ 1.0 (sampling from the model's own distribution). All models use their chat template and the same prompt; OLMo-3-1025-7B ships without one, so it gets its SFT checkpoint's template. Completions are capped at 4,096 tokens (8,192 for m1, whose long reasoning otherwise truncates), and generation stops at the end-of-turn token. The 32B models are sampled at $T$=0.7 only; the OLMo 3.1 32B SFT and DPO checkpoints, stored in float32, are run in bfloat16 like the other stages. We take the last valid JSON list in each completion. Unparsable completions count as incorrect; this affects mainly base models, so base-model pass@$k$ is a lower bound at large $k$, while post-trained gains at small $k$ are partly format. §4.3 reports format-controlled comparisons.

| Model | Parent | Post-training | Role |
|---|---|---|---|
| Qwen2.5-7B | n/a | none (pretrained) | base |
| Qwen2.5-7B-Instruct | Qwen2.5-7B | general SFT + RLHF | instruct; medical parent |
| AlphaMed-7B [17] | Qwen2.5-7B-Instruct | rule-based RL | medical |
| HuatuoGPT-o1-7B [1] | Qwen2.5-7B-Instruct | SFT + PPO (verifier) | medical |
| m1-7B-23K [9] | Qwen2.5-7B-Instruct | SFT distillation | medical |
| Qwen3-8B-Base | n/a | none (pretrained) | base |
| Qwen3-8B | Qwen3-8B-Base | general post-training | instruct |
| Qwen2.5-32B | n/a | none (pretrained) | base |
| Qwen2.5-32B-Instruct | Qwen2.5-32B | general SFT + RLHF | instruct |
| OLMo-3-1025-7B [28] | n/a | none (pretrained) | base |
| OLMo-3-7B-Instruct-SFT | OLMo-3-1025-7B | SFT | stage 1 |
| OLMo-3-7B-Instruct-DPO | OLMo-3 SFT | DPO | stage 2 |
| OLMo-3-7B-Instruct | OLMo-3 DPO | RLVR | stage 3 (instruct) |
| OLMo-3-7B-RL-Zero-General | OLMo-3-1025-7B | RL (LM judge) | RL from base |
| OLMo-3-1125-32B | n/a | none (pretrained) | base |
| OLMo-3.1-32B-Instruct-SFT | OLMo-3-1125-32B | SFT | stage 1 |
| OLMo-3.1-32B-Instruct-DPO | OLMo-3.1 SFT | DPO | stage 2 |
| OLMo-3.1-32B-Instruct | OLMo-3.1 DPO | RLVR | stage 3 (instruct) |

Table 1: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

**Metrics.** *pass@k*: the unbiased estimator of Chen et al. [2] on the top-1 answer of each sample. *Distinct* and *entropy*: the number of distinct top-1 diagnoses across a case's 32 samples, and the entropy of their empirical distribution, averaged over cases. On DDXPlus, $TV$: total-variation distance between this empirical distribution and the reference differential. *Coverage@k*: the expected share of the reference differential's probability mass covered by the distinct top-1 answers in $k$ random samples. *Calibration*: expected calibration error (10 bins) of the stated top-1 probability against top-1 correctness. These metrics describe the *sampled* readout. §4.6 defines the corresponding metrics for the *stated* readout, the list in a single completion. Differences come with 95% paired-bootstrap intervals over cases (2,000 resamples) and are computed from unrounded values. We report many intervals without multiplicity correction; we treat a non-significant difference as "no detectable change", with the interval bounding its size, not as equivalence.

# 4 Results

## 4.1 General post-training trades coverage for top-1 accuracy

Table 2 and Figure 1 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at $T$=1 (intervals $[+25.3, +29.6]$ and $[+20.7, +24.4]$). The advantage shrinks with $k$ and reverses: the curves cross at $k$≈4–8, and at $k$=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+7.8, +12.8]$). The pattern is the same at $T$=0.7 for Qwen3, crossing at $k$=4 with a base lead of 12.2 points at $k$=32 ($[+9.9, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at $T$=0.7 behaves the same way: +12.6 points at $k$=1 $[+10.7, +14.5]$ and a base lead of 9.7 at $k$=32 $[+7.3, +12.0]$.

| Pair | Model | pass@1 | pass@32 | Distinct | Entropy | TV↓ | Cov@32 | ECE↓ |
|---|---|---|---|---|---|---|---|---|
| Qwen3-8B, $T$=1 | base | 0.282 | **0.789** | **6.60** | **1.31** | **0.712** | **0.407** | 0.214 |
| | post-trained | **0.556** | 0.643 | 1.53 | 0.18 | 0.783 | 0.236 | **0.061** |
| Qwen2.5-7B, $T$=1 | base | 0.239 | **0.766** | **8.23** | **1.58** | **0.715** | **0.422** | 0.303 |
| | instruct | **0.464** | 0.663 | 2.65 | 0.49 | 0.761 | 0.290 | **0.137** |
| Qwen3-8B, $T$=0.7 | base | 0.388 | **0.730** | **3.76** | **0.78** | **0.750** | **0.322** | 0.121 |
| | post-trained | **0.558** | 0.607 | 1.27 | 0.11 | 0.793 | 0.217 | **0.063** |

Table 2: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

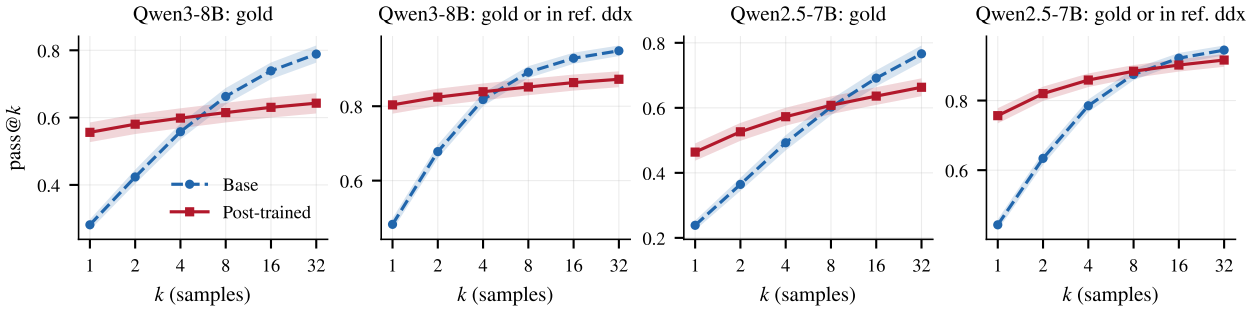



Figure 1: pass@$k$ on DDXPlus at $T$=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small $k$ and loses at large $k$.

### 4.2 The sampled differential collapses, while top-1 calibration improves

The gap at large $k$ reflects a collapsed *sampled* differential. Across 32 samples, the post-trained Qwen models name 1.3–2.6 distinct diagnoses per case, against 3.8–8.3 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.58 for base models and 1.87 for the reference differential (Table 2). Sampling at $T$=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at $T$=0.7, 0.18 at $T$=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the reference differential: 0.24 vs. 0.41 of its probability mass at $k$=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). On every single-answer criterion (accuracy, small-$k$ pass@$k$, calibration), the post-trained model therefore looks strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.6).

**Which diagnoses drop out?** For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 3), and the severe tier (DDXPlus severity 1–2, the dataset's label rather than a clinical consensus list) is not spared. This metric needs a chance reference, though: a guesser that picks uniformly among the 49 conditions scores about 0.48 on it, and one that draws from the average reference differential 0.56–

| Comparison | Considered@32 (base → post-trained) Severe (1–2) | Moderate (3) | Mild (4–5) |
|---|---|---|---|
| Qwen3-8B, $T$=1 | 0.45 → 0.27 | 0.50 → 0.32 | 0.45 → 0.20 |
| | −18.4 [−20.8, −15.9] | −18.1 [−20.7, −15.7] | −24.2 [−27.0, −21.4] |
| Qwen2.5-7B, $T$=1 | 0.51 → 0.34 | 0.50 → 0.37 | 0.38 → 0.23 |
| | −17.0 [−19.7, −14.2] | −12.1 [−14.7, −9.7] | −15.0 [−17.4, −12.6] |
| Qwen3-8B, $T$=0.7 | 0.34 → 0.24 | 0.44 → 0.29 | 0.31 → 0.18 |
| | −9.8 [−11.8, −7.9] | −15.8 [−18.2, −13.5] | −13.4 [−15.8, −11.1] |
| Qwen2.5-7B, $T$=0.7 | 0.40 → 0.28 | 0.41 → 0.32 | 0.28 → 0.17 |
| | −12.4 [−14.7, −10.2] | −9.1 [−11.2, −6.7] | −10.3 [−12.3, −8.3] |
| Instruct → AlphaMed, $T$=0.7 | 0.28 → 0.32 | 0.32 → 0.34 | 0.17 → 0.21 |
| | +4.3 [+2.0, +6.6] | +2.2 [+0.4, +4.0] | +3.9 [+2.4, +5.4] |
| Instruct → HuatuoGPT-o1, $T$=0.7 | 0.28 → 0.36 | 0.32 → 0.37 | 0.17 → 0.27 |
| | +8.4 [+6.1, +10.8] | +6.0 [+4.0, +7.9] | +9.5 [+7.5, +11.5] |
| Instruct → m1, $T$=0.7 | 0.28 → 0.41 | 0.32 → 0.41 | 0.17 → 0.29 |
| | +13.5 [+10.8, +16.2] | +9.7 [+7.2, +12.2] | +11.4 [+9.2, +13.6] |
| Base → AlphaMed (base-RL), $T$=0.7 | 0.40 → 0.39 | 0.41 → 0.40 | 0.28 → 0.27 |
| | −0.7 [−3.0, +1.7] | −0.4 [−2.6, +1.7] | −0.4 [−2.4, +1.7] |
| Base → `acc` GRPO (8B), $T$=0.7 | 0.34 → 0.31 | 0.44 → 0.42 | 0.31 → 0.27 |
| | −2.8 [−4.4, −1.3] | −2.1 [−3.7, −0.6] | −4.3 [−6.2, −2.5] |
| Base → `gm` GRPO (8B), $T$=0.7 | 0.34 → 0.32 | 0.44 → 0.43 | 0.31 → 0.28 |
| | −1.5 [−3.2, +0.1] | −1.3 [−3.0, +0.3] | −3.0 [−4.8, −1.2] |

Table 3: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the can't-miss tier; medical post-training raises it; our GRPO from the base model loses at most 4 points, and with the group reward the severe-tier loss is not significant.

0.61, above every model, base models included. Considered@32 therefore measures the breadth of the sampled answers, not clinical judgement; we assess can't-miss conditions on the stated list instead (§4.9). For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at $T$=1. The differences are −18.4 [−20.8, −15.9] and −17.0 [−19.7, −14.2] points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives, which is a defensible triage bias but not a differential.

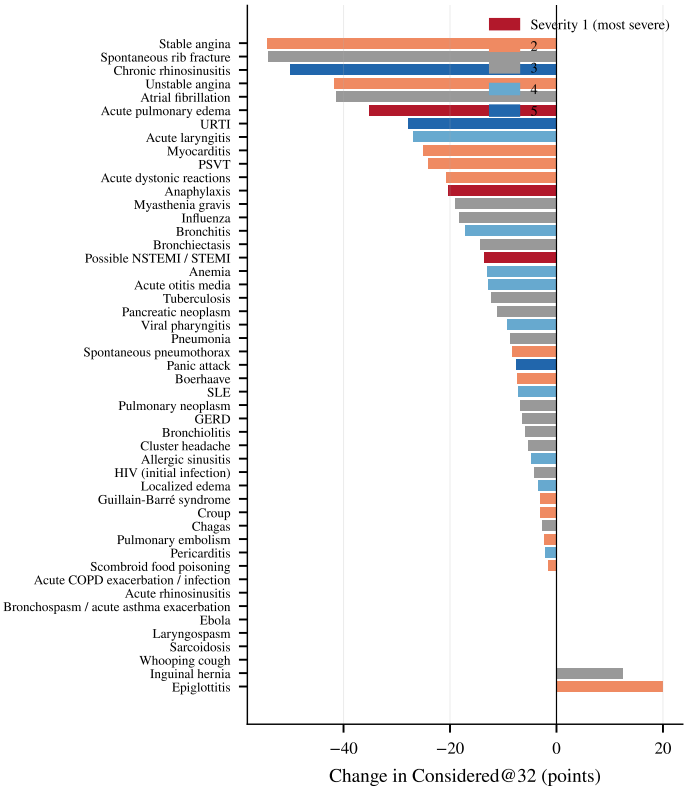



Figure 2: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model ($T$=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability $\geq 0.1$ are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

### 4.3 Robustness to prompt and output format

**Prompt.** We repeat the Qwen3-8B pair with two other prompts (Table 4). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern: the post-trained model gains +23.0 [+20.9, +25.1] points at $k$=1 and loses −7.1 [−9.3, −5.0] at $k$=32 ($T$=0.7; at $T$=1, +34.2 and −8.7 [−11.0, −6.4]), with 1.3–1.5 distinct answers against 3.7–5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every $k$ at $T$=0.7 (+4.7 [+2.9, +6.6] at $k$=32) and ties it at $T$=1 (0.0 [−2.1, +1.9]). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at $k$=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, as in most clinical decision-support deployments, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§4.11).

**Output format.** Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at $T$=0.7 and 25% at $T$=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large $k$ but inflates the post-trained model's gain at small $k$. Among valid samples only, the post-trained models still gain +12.0 [+9.8, +14.1] (Qwen3,

| Prompt | $T$ | pass@1 | | pass@32 | | Distinct | |
|---|---|---|---|---|---|---|---|
| | | Base | Post | Base | Post | Base | Post |
| default | 0.7 | 0.388 | **0.558** | **0.730** | 0.607 | 3.76 | 1.27 |
| default | 1.0 | 0.282 | **0.556** | **0.789** | 0.643 | 6.60 | 1.53 |
| plain | 0.7 | 0.352 | **0.583** | **0.727** | 0.655 | 3.72 | 1.28 |
| plain | 1.0 | 0.244 | **0.586** | **0.792** | 0.705 | 5.57 | 1.54 |
| CoT | 0.7 | 0.409 | **0.652** | 0.788 | **0.835** | 3.89 | 2.08 |
| CoT | 1.0 | 0.275 | **0.642** | 0.837 | 0.837 | 6.31 | 2.24 |

Table 4: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at $k$=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

$T$=0.7), +18.7 [+16.6, +20.7] (Qwen3, $T$=1), +12.2 [+10.4, +14.0] and +16.8 [+15.0, +18.5] points (Qwen2.5) at $k$=1, so a quarter to a third of the raw gain at $T$=1 is format. On the cases where every sample of both models is valid, the large-$k$ loss persists: −15.2 [−21.9, −8.6] at $k$=32 for Qwen3 (105 cases) and −10.2 [−12.7, −7.6] for Qwen2.5 (748 cases), both at $T$=0.7. (At $T$=1 almost no case has 32 valid base samples.) The reported large-$k$ losses are thus lower bounds.

### 4.4 Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 [28] releases every stage of its 7B instruct pipeline from one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 5 gives the results.

The coverage loss replicates in this third family. At $T$=0.7, every post-trained stage is worse than the base at $k$=32: by −14.6 [−17.5, −11.7] points after SFT, −9.2 [−12.1, −6.1] after DPO, −28.3 [−31.5, −25.0] after RLVR, and −13.3 [−16.1, −10.5] for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large $k$ (+5.4 [+2.9, +8.0] points at $k$=32, and +10.3 [+9.0, +11.6] at $k$=1) while narrowing the differential further (−0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses −19.1 [−21.8, −16.3] points at $k$=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at $k$=1 (−5.9 [−7.2, −4.6]). It drops severe-tier Considered@32 to 0.25, a loss of −16.1 [−18.6, −13.5] points relative to the base, against −7.4 for the DPO model. At $T$=1 the pattern is the same (RLVR vs. DPO: −12.7 [−15.4, −9.9] at $k$=32).

RL applied directly to the base narrows the differential less. At $T$=1, RL-Zero shows no detectable loss of the base model's pass@32 (−1.8 [−4.5, +0.8], so at most 4.5 points) and 6.28 of its 7.53 distinct answers, and it loses 4.0 [1.2, 6.9] points of severe-tier consideration, against 14.4 for the full instruct pipeline. At $T$=0.7 it still loses 13.3 points at $k$=32, about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at $T$=0.7 and 47% at $T$=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid check of §4.3 is not possible here. Among valid samples, the post-trained models' gains at $k$=1 are small (−1.9 to +4.0 points

| $T$ | Stage | pass@1 | pass@32 | Distinct | Entropy | Cov@32 | Severe C@32 |
|---|---|---|---|---|---|---|---|
| 0.7 | Base | 0.178 | **0.705** | **4.93** | **1.16** | **0.351** | **0.412** |
| | SFT | 0.222 | 0.559 | 4.01 | 0.87 | 0.257 | 0.373 |
| | SFT+DPO | **0.324** | 0.613 | 3.30 | 0.63 | 0.272 | 0.338 |
| | SFT+DPO+RLVR (Instruct) | 0.266 | 0.422 | 2.37 | 0.42 | 0.195 | 0.251 |
| | RL-Zero (RL from base) | 0.295 | 0.572 | 4.06 | 0.84 | 0.286 | 0.326 |
| 1.0 | Base | 0.128 | **0.713** | **7.53** | **1.66** | **0.398** | **0.469** |
| | SFT | 0.191 | 0.634 | 6.32 | 1.29 | 0.321 | 0.426 |
| | SFT+DPO | **0.313** | 0.688 | 4.48 | 0.87 | 0.312 | 0.410 |
| | SFT+DPO+RLVR (Instruct) | 0.272 | 0.561 | 3.58 | 0.68 | 0.259 | 0.325 |
| | RL-Zero (RL from base) | 0.279 | 0.695 | 6.28 | 1.21 | 0.367 | 0.429 |

Table 5: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 3. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at $T$=1.

at $T$=0.7, +1.3 to +8.1 at $T$=1), so most of OLMo's raw gain at $k$=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across three families, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§4.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero, AlphaMed's base-initialized checkpoint, and our GRPO runs in §5) narrows the differential less. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at $k$=32 ($-2.0$ $[-3.0, -1.1]$ points after SFT, $-1.5$ after DPO, $-1.4$ after RLVR; all significant), while RL-Zero does not ($-0.6$ $[-1.4, +0.2]$), from a base model at 4.5% (§4.12).

### 4.5 Replication at 32B

We repeat the DDXPlus comparisons at $T$=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 6). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 $[2.8, 7.1]$ points at pass@32 for Qwen2.5-32B and 4.0 $[2.0, 5.9]$ for OLMo 3.1 32B, halves the number of distinct answers (3.78 $\rightarrow$ 1.68 and 4.32 $\rightarrow$ 2.15), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid ($-4.8$ $[-7.8, -1.7]$); the OLMo 32B base gives invalid answers on 27% of samples. The stage attribution differs from 7B. In OLMo 3.1 32B the largest narrowing is SFT (pass@32 $-12.1$ $[-14.4, -9.9]$, distinct answers $-1.39$, severe tier $-11.9$); DPO recovers most of it (+10.9 against SFT), and the final RL step is small ($-2.8$ $[-4.3, -1.1]$ at $k$=32, $-0.14$ distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. The stated readout improves in both 32B families (§4.6), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (the latter n.s.).

| Model | pass@1 | pass@32 | Distinct | Cov@32 | Severe C@32 | TV one list↓ | TV samples↓ | Vote@32 gain |
|---|---|---|---|---|---|---|---|---|
| Qwen2.5-32B (base) | 0.500 | **0.795** | **3.78** | **0.352** | **0.35** | 0.726 | **0.726** | +7.0 |
| Qwen2.5-32B-Instruct | **0.604** | 0.746 | 1.68 | 0.276 | 0.29 | **0.674** | 0.759 | +1.9 |
| OLMo-3-1125-32B (base) | 0.326 | **0.796** | **4.32** | **0.347** | **0.39** | 0.746 | **0.742** | +8.4 |
| SFT | 0.461 | 0.674 | 2.94 | 0.265 | 0.27 | 0.736 | 0.776 | +3.6 |
| SFT+DPO | **0.584** | 0.784 | 2.29 | 0.293 | 0.32 | **0.694** | 0.755 | +2.1 |
| SFT+DPO+RLVR (Instruct) | 0.560 | 0.756 | 2.15 | 0.283 | 0.30 | 0.701 | 0.763 | +0.9 |

Table 6: 32B replication on DDXPlus ($T$=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.6). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.7). Bold: best value per family.

### 4.6 The stated differential does not collapse

Every completion also *states* a differential: up to five conditions with probabilities. We read it directly. For each valid sample we renormalize its stated probabilities over the conditions it names (uniform if the probabilities are missing or invalid) and compare this single list with the reference differential: its total-variation distance to the reference, the reference mass of the conditions it names, and, for each severe condition with reference probability at least 0.1, whether the list names it. For the sampled readout we use the empirical distribution of top-1 answers over the valid samples and, for the same budget of five names, the reference mass covered by the distinct top answers of five random samples. Two oracles bound what each format can achieve: the reference's own top five stated as a list, and as many draws from the reference as the model has valid samples (Table 7, Figure 3).

For the Qwen models the two readouts move in opposite directions under post-training. In both families and at both temperatures, the post-trained model's single list is closer to the reference than the base model's (TV −0.043 to −0.086, all intervals exclude zero), covers more of its mass (+6.1 to +10.8 points), and names severe conditions more often (+9.1 to +13.4 points). Its sampled top answers move farther from the reference (+0.029 to +0.071), and five of them cover less (−1.9 to −6.0 points). Within each post-trained model the list beats its own samples by 0.04–0.09 in TV; within each base model the gap is about zero or favors the samples. The CoT prompt keeps the list's advantage (TV −0.042, severe naming +8.1 points) and removes the loss in the samples (+0.002, n.s.). This quantifies the observation of §4.2 that conditions dropping out of the top answer remain lower in the list.

At 32B the pattern holds in both families (Table 6): the post-trained lists are closer to the reference (TV −0.053 for Qwen2.5-32B-Instruct, −0.052 and −0.045 for the OLMo 3.1 DPO and Instruct models) and cover 5.6–6.7 points more of it, while the samples move away (+0.014 to +0.034). Within every post-trained 32B model the list beats its own samples by 0.04–0.09 in TV, against about zero for both base models. OLMo 3 7B is the exception with the five-item prompt. Its DPO and Instruct models' lists are no closer to the reference than the base model's (TV +0.000 and +0.028 at $T$=0.7; −0.026 and −0.005 at $T$=1) and cover at most 5 points more of it; only the naming of severe conditions improves (+6.1 to +12.8 points). Asked for a probability on every condition, its Instruct model's stated distribution is closer to the reference than the base model's (−0.032 [−0.038, −0.027]; §4.8), so the exception is specific to the short list. RL-Zero's lists improve slightly (TV −0.020 and −0.039). The medical descendants change little relative to their parent (TV −0.012 for AlphaMed, −0.014 for HuatuoGPT-o1, +0.017 for m1). In absolute terms every model is far from the reference: a single list covers 0.22–0.41 of the reference mass, against 0.74 for the reference's own top five.

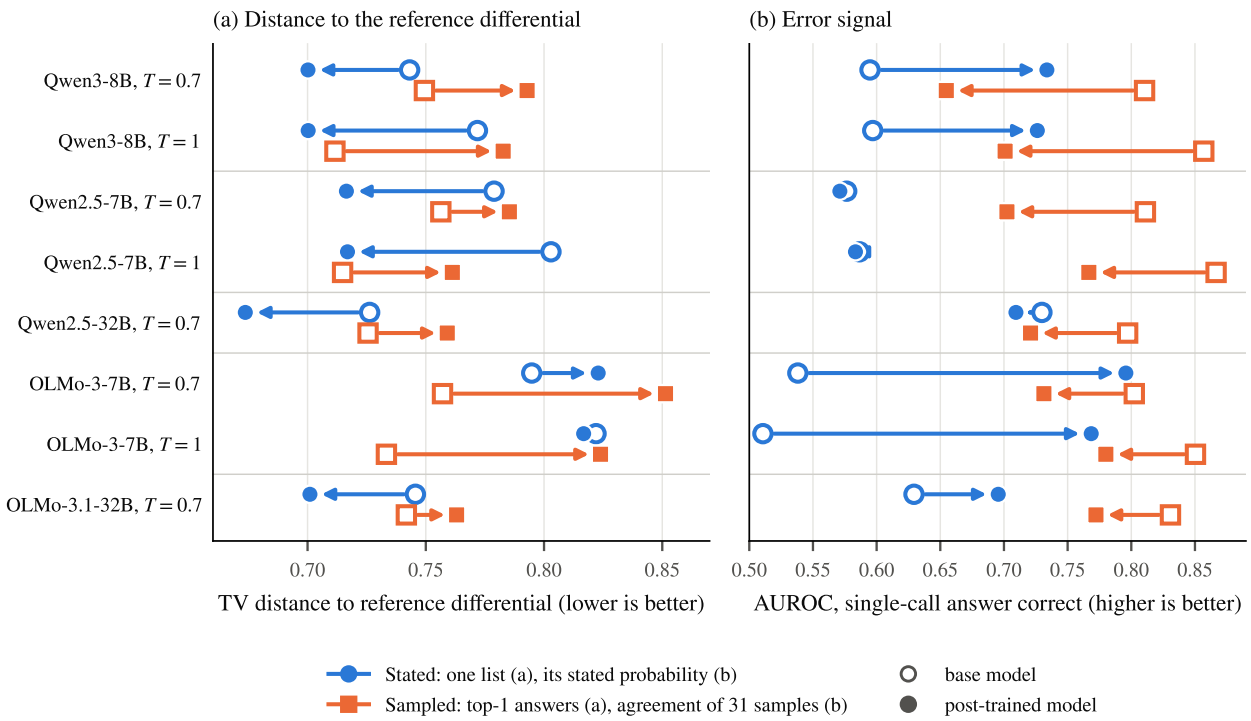



Figure 3: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; 32B models at $T$=0.7 only). (a) TV distance to the reference differential of one stated list (blue circles) and of the sampled top-1 answers (orange squares). (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

### 4.7 Consequences for sampling-based methods

We ask what the gap does to two common uses of repeated samples: aggregating them into an answer, and using their agreement as an uncertainty signal (Table 8). The single-call answer is the top answer of a model's first valid sample.

**Aggregation.** Majority voting over 32 samples (self-consistency; 30) raises base models' accuracy by 3.4–5.2 points at $T$=0.7 and 10.4–14.2 at $T$=1. For the general post-trained models it adds at most 0.8 points at $T$=0.7 (Qwen3-8B +0.3, Qwen2.5-7B-Instruct +0.8, OLMo-3-Instruct −0.2; all n.s.) and 0.1–4.4 at $T$=1. Models that reason before answering still benefit (HuatuoGPT-o1 +4.8, m1 +3.8, Qwen3-8B with the CoT prompt +4.5). Averaging the stated lists instead of voting changes accuracy by at most about 2 points for any model. For a top-three differential, the stated list is the better source after post-training. For Qwen3-8B, the top three of one list contains the true pathology 13.7 [+11.4, +15.9] points more often than the three most frequent answers in 32 samples ($T$=1: +10.0), and for Qwen2.5-7B-Instruct 7.3 [+5.3, +9.6] points more often ($T$=1: +2.0, n.s.). For every base model the 32 samples win, by 7.3–13.6 points at $T$=0.7. OLMo-3-Instruct is in between (+1.9, n.s.; $T$=1: −4.7).

**Uncertainty.** We score how well each signal ranks the correctness of the single-call answer (AUROC). The stated probability of that answer needs no extra calls; *agreement*, the share of the

| Pair | Model | TV to reference↓ One list | TV to reference↓ Samples | Reference mass covered One list | Reference mass covered 5 samples | Severe named One list |
|---|---|---|---|---|---|---|
| Qwen3-8B, $T$=0.7 | base | 0.743 | **0.749** | 0.335 | **0.245** | 0.347 |
| | post-trained | **0.700** | 0.793 | **0.396** | 0.207 | **0.437** |
| Qwen3-8B, $T$=1 | base | 0.772 | **0.712** | 0.293 | **0.275** | 0.312 |
| | post-trained | **0.700** | 0.783 | **0.396** | 0.215 | **0.440** |
| Qwen2.5-7B, $T$=0.7 | base | 0.779 | **0.756** | 0.291 | **0.233** | 0.345 |
| | instruct | **0.716** | 0.785 | **0.364** | 0.214 | **0.440** |
| Qwen2.5-7B, $T$=1 | base | 0.803 | **0.715** | 0.255 | **0.257** | 0.304 |
| | instruct | **0.717** | 0.761 | **0.363** | 0.234 | **0.438** |
| OLMo 3 7B, $T$=0.7 | base | 0.795 | **0.757** | 0.268 | **0.237** | 0.256 |
| | SFT+DPO | 0.795 | 0.804 | 0.277 | 0.193 | **0.378** |
| | Instruct (RLVR) | 0.823 | 0.851 | 0.257 | 0.150 | 0.317 |
| | RL-Zero | **0.775** | 0.793 | **0.295** | 0.197 | 0.274 |
| Oracle | reference | 0.265 | 0.18–0.25 | 0.735 | 0.563 | 0.975 |

Table 7: Stated vs. sampled differentials on DDXPlus (980 cases, valid samples only). *One list*: a single sample's stated differential, renormalized over the conditions it names (mean over samples). *Samples*: the empirical distribution of top-1 answers over the 32 samples; *5 samples*: reference mass of the distinct top answers in five random samples, the same five-name budget as one list. *Severe named*: share of (case, severe condition with reference probability $\geq 0.1$) pairs that one list names. Bold marks the better model of each pair (OLMo: of the block). Oracle: the reference's own top five as a list, and draws from the reference as samples. Post-training improves the Qwen lists while worsening their samples; OLMo's lists do not improve.

model's other samples with the same top answer, uses 4 or 31 more. For base models agreement is far better: 0.80–0.81 with 31 extra samples, against 0.54–0.60 for their nearly uninformative stated probability. Post-training weakens agreement in every family (Qwen3 0.810 → 0.655, Qwen2.5 0.811 → 0.702, OLMo 0.802 → 0.731), consistent with Liu [18], and moves the comparison toward the stated probability. For Qwen3-8B (0.734) and OLMo-3-Instruct (0.796), the stated probability is better than agreement over 32 samples (+0.079 [+0.046, +0.110] and +0.064 [+0.027, +0.101]); at $T$=1 the two are statistically tied. For Qwen2.5-7B-Instruct (0.571 vs. 0.702) and all three medical models, agreement remains better. These AUROCs are computed on each model's own answers, so they compare signals within a model, not models with each other.

**Ambiguity.** The reference differential also says how ambiguous each case is. We correlate each uncertainty signal with the entropy of the case's reference differential. For every model and signal the Spearman correlation lies between −0.10 and +0.14, except AlphaMed's stated probability and stated entropy (+0.30 and +0.32). How uncertain a model is about a case, by either readout, says almost nothing about how ambiguous the case is.

**At 32B.** The same shifts appear with smaller margins. Voting gains fall from 7.0 and 8.4 points for the base models to 1.9 and 0.9 (the latter n.s.) for the Instruct models, and agreement weakens as an error signal (Qwen 0.797 → 0.721, OLMo 0.831 → 0.772). For Qwen2.5-32B-Instruct the top three of one list beats the three most frequent of 32 sampled answers by 5.6 [3.6, 7.7] points, but its stated probability only ties agreement over 32 samples (−0.011, n.s.; it beats agreement over five, +0.047); for OLMo 3.1 32B Instruct agreement stays better (−0.077).

| Model | Single-call accuracy | Vote@32 gain | Top-3: list − 32 samples | AUROC, single-call answer correct | | |
|---|---|---|---|---|---|---|
| | | | | Stated $p$ | Agree@5 | Agree@32 |
| Qwen3-8B-Base | 0.456 | +3.4 | −7.3 | 0.595 | 0.772 | **0.810** |
| Qwen3-8B | 0.556 | +0.3 | +13.7 | **0.734** | 0.613 | 0.655 |
| Qwen3-8B (CoT prompt) | 0.648 | +4.5 | +0.0 | 0.718 | 0.725 | **0.778** |
| Qwen2.5-7B | 0.371 | +5.2 | −8.9 | 0.576 | 0.764 | **0.811** |
| Qwen2.5-7B-Instruct | 0.480 | +0.8 | +7.3 | 0.571 | 0.653 | **0.702** |
| OLMo-3 base | 0.291 | +4.0 | −13.6 | 0.538 | 0.761 | **0.802** |
| OLMo-3 SFT+DPO | 0.327 | −0.4 | −7.8 | 0.738 | 0.702 | **0.751** |
| OLMo-3-Instruct | 0.267 | −0.2 | +1.9 | **0.796** | 0.668 | 0.731 |
| AlphaMed-7B | 0.528 | +0.9 | +0.6 | 0.673 | 0.663 | **0.715** |
| HuatuoGPT-o1-7B | 0.532 | +4.8 | −0.4 | 0.590 | 0.715 | **0.775** |
| m1-7B | 0.495 | +3.8 | −4.6 | 0.680 | 0.736 | **0.782** |

Table 8: Aggregation and uncertainty on DDXPlus ($T$=0.7, 980 cases). Single-call answer: top-1 of the first valid sample. Vote@32 gain: accuracy of the majority top-1 over all valid samples minus single-call accuracy, in points. Top-3: rate at which one stated list's top three contains the true pathology, minus the rate for the three most frequent sampled answers (points). AUROC for the single-call answer being correct: its stated probability (no extra calls) vs. agreement of 4 or 31 further samples; bold marks the best signal. Post-trained models gain little from voting; for some of them the stated probability becomes the better error signal.

After post-training, then, repeated sampling adds little: voting no longer improves accuracy and agreement carries less signal, while one stated list carries as much of the differential or more. Where the crossover falls depends on the model, so neither readout can be assumed to be the informative one without checking.

### 4.8 Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at $T$=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 9). The results match the short list. After post-training the stated distribution is closer to the reference (TV −0.052 for Qwen3-8B, −0.067 for Qwen2.5-7B-Instruct, −0.032 for OLMo-3-7B-Instruct) while the samples move away (+0.012 to +0.038). Qwen3-8B's stated probability again beats agreement over 32 samples (+0.069 [+0.033, +0.104]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) but not for the other two models.

### 4.9 Ruling out format, accuracy and floor effects

Table 10 tests simpler explanations of the stated-list result. *Format.* The general models' post-trained lists and base-model lists at $T$=0.7 are nearly all complete (81–99% name five on-list conditions with probabilities; Qwen base models at $T$=1, 61–62%; the medical models, 61–80%). Restricting to complete lists leaves the gaps unchanged (Qwen2.5-7B: 0.779 vs. 0.719). *Accuracy.* A more accurate model puts more stated mass on the true pathology, which the reference also favors. Within samples whose top answer is correct, and within those whose top answer is wrong,

| Model | TV to reference↓ One distribution | Samples | AUROC, single-call answer correct Stated $p$ | Agree@5 | Agree@32 | Top-3: one dist. − 32 samples |
|---|---|---|---|---|---|---|
| Qwen3-8B-Base | 0.733 | **0.752** | 0.647 | 0.809 | **0.845** | −6.7 |
| Qwen3-8B | **0.681** | 0.790 | **0.717** | 0.609 | 0.647 | +14.6 |
| Qwen2.5-7B | 0.769 | **0.758** | 0.688 | 0.792 | **0.827** | −10.2 |
| Qwen2.5-7B-Instruct | **0.702** | 0.770 | 0.674 | 0.715 | **0.748** | +1.6 |
| OLMo-3-1025-7B | 0.807 | **0.766** | 0.610 | 0.744 | **0.786** | −15.3 |
| OLMo-3-7B-Instruct | **0.775** | 0.787 | 0.749 | 0.705 | **0.759** | −9.5 |

Table 9: Full-distribution prompt on DDXPlus ($T$=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

| Model ($T$=0.7) | Complete lists | TV of one list↓ complete only | top-1 right | top-1 wrong | Excess TV over floor↓ One list | Samples | Severe named ref. < 0.1 | pass@32 list-sampled |
|---|---|---|---|---|---|---|---|---|
| Qwen3-8B-Base | 0.90 | 0.743 | 0.663 | 0.806 | 0.479 | **0.557** | 0.752 | 0.780 |
| Qwen3-8B | 0.99 | **0.701** | **0.646** | **0.769** | **0.435** | 0.613 | **0.680** | **0.811** |
| Qwen2.5-7B | 0.95 | 0.779 | 0.689 | 0.828 | 0.514 | **0.575** | 0.823 | 0.772 |
| Qwen2.5-7B-Instruct | 0.91 | **0.719** | **0.655** | **0.773** | **0.452** | 0.605 | **0.796** | **0.802** |
| Qwen2.5-32B | 0.96 | 0.725 | 0.661 | 0.795 | 0.462 | **0.543** | 0.735 | 0.850 |
| Qwen2.5-32B-Instruct | 0.98 | **0.674** | **0.639** | **0.726** | **0.409** | 0.579 | **0.645** | **0.880** |
| OLMo-3-1025-7B | 0.81 | **0.794** | 0.687 | **0.837** | **0.524** | **0.522** | **0.747** | **0.719** |
| OLMo-3-7B-Instruct | 0.95 | 0.823 | **0.659** | 0.882 | 0.558 | 0.671 | 0.779 | 0.577 |
| OLMo-3-1125-32B | 0.89 | 0.744 | 0.662 | 0.812 | 0.481 | **0.532** | 0.706 | 0.839 |
| OLMo-3.1-32B-Instruct | 0.99 | **0.701** | **0.644** | **0.774** | **0.436** | 0.583 | **0.634** | **0.853** |

Table 10: Robustness of the stated-list result (DDXPlus, $T$=0.7, 980 cases, valid samples). Complete lists: share naming five on-list conditions with valid probabilities. TV of one list on complete lists only, and on samples whose top answer is right or wrong. Excess TV: TV minus the expected TV of the matching oracle (the reference's top five for one list; as many reference draws as the model has valid samples for the samples). Severe named, ref. < 0.1: share of named severe conditions that the reference gives less than 0.1 (over-naming). pass@32 list-sampled: top answers drawn from each sample's stated list. Bold: better model of each pair. A uniform guesser scores about 0.48 at pass@32.

the post-trained Qwen list is still closer to the reference (−0.018 to −0.034 and −0.036 to −0.079 in TV). *Floor.* One list can do no better than the reference's own top five, and a finite set of samples no better than as many draws from the reference. Measured as excess over these floors, post-training lowers the list's TV (Qwen3-8B 0.479 → 0.435) and raises the samples' (0.557 → 0.613). *Over-naming.* Post-trained lists name fewer conditions that the reference gives less than 1% (Qwen3-8B 0.59 → 0.55 of named conditions) and fewer severe conditions it gives less than 0.1 (Qwen2.5-32B 0.74 → 0.65), so the gain in naming severe conditions is not over-triage. *Sampling from the list.* Replacing each sample's top answer with a draw from its own stated list restores the coverage the samples lost: post-trained models' pass@32 then exceeds their base models' sampled pass@32 in every comparison except OLMo 3 7B (Qwen3-8B 0.811 vs. 0.730; Qwen2.5-32B-Instruct 0.880 vs. 0.795; OLMo 3.1 32B Instruct 0.853 vs. 0.796), against about 0.48 for a uniform guesser. OLMo 3 7B remains the exception throughout: its Instruct list is closer to the reference when its top answer is correct but much farther when it is wrong.

### 4.10 Which readout to trust: a held-out audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell. We pool 56 model–setting combinations scored on DDXPlus: the base/post-trained pairs at both temperatures and with three prompts, the OLMo stages, the 32B models, the medical models, our GRPO runs, and the full-distribution prompt. They share weights, prompts and temperatures, so we treat the correlations below as descriptive. The stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), with the stated probability's own AUROC (0.82; the gap contains this term, so part of this is by construction), and negatively with the number of distinct answers ($-0.74$). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. The simplest rule, always using agreement, is right for 66% of the combinations (54% of the 39 post-trained ones) and loses 0.025 AUROC on average (0.032). An audit does better: choose the signal with the higher AUROC on 100 random labeled cases, using five samples per case, and evaluate the choice on the other 880. Over 500 draws per combination it picks the better held-out signal 86% of the time (82% for post-trained models) and loses 0.005 AUROC (0.007); with 50 cases, 82% and 0.010; with 200, 90% and 0.003. It is reliable when the two signals differ by more than 0.05 AUROC (96% of draws, 39 combinations) and close to a coin flip when they differ by less than 0.02 (53%, 5 combinations), where the choice matters least. The audit's main cost is gold labels for 100 cases from the target population, and we validate it only within DDXPlus.

### 4.11 Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every $k$ (Figure 4): +3.9 points at $k$=1 ($[+2.2, +5.5]$), widening to +8.4 at $k$=32 ($[+6.0, +10.6]$). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.29 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 3). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 11). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 $[+11.3, +16.0]$ and 18.1 $[+15.5, +20.7]$ points. m1 gains almost nothing at $k$=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). Medical post-training on an instruction-tuned model thus never adds to the collapse, and reasoning distillation partially reverses it.

**Medical RL from the base model.** Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper [17] describes

| Model | Post-training | pass@1 | pass@32 | Distinct | Entropy | Cov@32 | ECE |
|---|---|---|---|---|---|---|---|
| Qwen2.5-7B-Instruct | (parent) | 0.478 | 0.598 | 1.82 | 0.30 | 0.239 | **0.106** |
| AlphaMed-7B | rule-based RL | 0.516 | 0.682 | 1.87 | 0.28 | 0.266 | 0.121 |
| HuatuoGPT-o1-7B | SFT + PPO | **0.536** | 0.735 | 2.49 | 0.44 | 0.306 | 0.168 |
| m1-7B-23K | SFT distillation | 0.486 | 0.779 | 3.29 | 0.62 | 0.328 | 0.206 |
| Qwen2.5-7B (base) | (parent) | 0.352 | 0.695 | **4.91** | **1.00** | **0.335** | 0.291 |
| AlphaMed-7B (base-RL) | rule-based RL | 0.495 | **0.790** | 3.14 | 0.62 | 0.320 | 0.202 |

Table 11: Medical models on DDXPlus (980 cases, 32 samples, $T$=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best pass@32 of any 7B model.

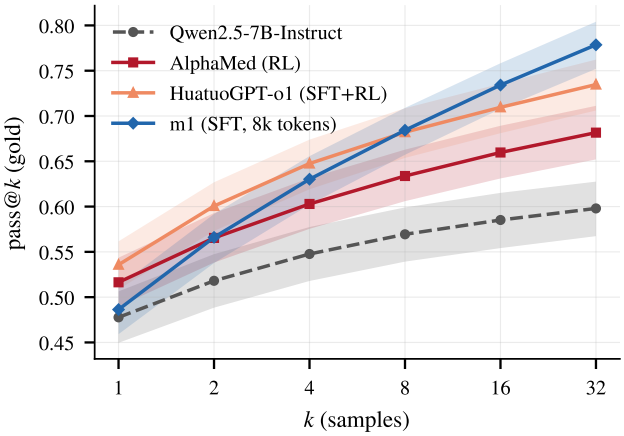



Figure 4: DDXPlus at $T$=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every $k$, and the gaps widen with $k$: there is no sharpening tax at the medical stage.

only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 11, lower block). It does not. RL from the base model improves on the base at every $k$ (+14.3 [+13.1, +15.7] points at $k$=1, +9.5 [+7.4, +11.6] at $k$=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 [0.4, 3.9] points worse at $k$=1 and 10.8 [+8.6, +13.2] points better at $k$=32. Its severe-tier Considered@32 shows no detectable change relative to the base (−0.7 [−3.0, +1.7] points), against −12.4 for instruction tuning and −8.1 for instruction tuning followed by AlphaMed's RL (Table 3; all well below the chance levels of §4.2). This is consistent with our own controlled GRPO-from-base result (§5): with a verifiable correctness reward, RL from a base model buys accuracy without the collapse, which is introduced by general instruction tuning and inherited by everything trained on top of it. OLMo 3 refines this (§4.4): within its instruct pipeline, the RL stage applied after SFT and DPO is the largest single narrowing.

| Model | ≤2022 ($n$=901) pass@1 | pass@32 | 2023–24 ($n$=348) pass@1 | pass@32 | 2025+ ($n$=152) pass@1 | pass@32 |
|---|---|---|---|---|---|---|
| Qwen2.5-7B (base) | 2.0 | 8.2 | 0.0 | 0.3 | 0.6 | 3.3 |
| Qwen2.5-7B-Instruct | 2.3 | 7.7 | 0.0 | 0.9 | 0.8 | 3.9 |
| AlphaMed-7B | 3.0 | 9.0 | 0.3 | 1.4 | 1.4 | 3.9 |
| AlphaMed-7B (base-RL) | 2.4 | 8.8 | 0.1 | 1.1 | 1.4 | 4.6 |
| HuatuoGPT-o1-7B | 3.1 | 8.4 | 0.3 | 1.4 | 1.3 | 3.9 |
| m1-7B | 2.6 | 9.5 | 0.3 | 1.7 | 1.0 | 3.9 |
| Qwen3-8B-Base | 2.1 | 8.3 | 0.1 | 0.6 | 1.1 | 3.9 |
| Qwen3-8B | 3.1 | 5.2 | 0.1 | 0.3 | 2.0 | 2.6 |

Table 12: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, $T$=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases; 2025+ cases fall in between.

### 4.12 Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 is 3.7–7.0%. Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67; at this accuracy ECE mostly restates mean confidence, so we read it as a rise in confidence without a matching rise in accuracy. Medical RL thus increases overconfidence exactly where the model is out of its depth. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident model we evaluate (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: AlphaMed beats its parent at every $k$ (+0.6 [+0.2, +0.9] points at $k$=1, +1.0 [+0.1, +2.1] at $k$=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ (+1.4 [+0.4, +2.5] at $k$=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 (+0.7 [+0.4, +1.1]) and trails it from $k$=8 on, by −2.2 [−3.1, −1.3] points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@$k$ difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§5) are indistinguishable from their base (all pass@$k$ differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published (Table 12). For every model, pass@32 is 5–10% on cases published up to 2022 ($n$=901), at most 1.7% on 2023–24 cases ($n$=348), and 2.6–4.6% on 2025+ cases ($n$=152). The pattern is not monotone, since 2025+ cases score above 2023–24 cases, so case mix also matters; the gap between older and recent cases is consistent with, but does not establish, memorization of older case reports.

| Model | Single-call correct | Vote@32 gain | Top-3: list − 32 samples | AUROC, single-call answer correct | | | Correct answers |
|---|---|---|---|---|---|---|---|
| | | | | Stated $p$ | Agree@5 | Agree@32 | |
| Qwen3-8B-Base | 0.074 | +4.6 | −7.0 | 0.567 | 0.768 | **0.830** | 104 |
| Qwen3-8B | 0.080 | +0.4 | +0.4 | 0.707 | 0.726 | **0.763** | 112 |
| Qwen2.5-7B | 0.062 | +3.8 | −5.4 | 0.598 | 0.788 | **0.845** | 87 |
| Qwen2.5-7B-Instruct | 0.056 | +1.5 | −3.6 | 0.717 | 0.661 | **0.745** | 78 |
| OLMo-3-1025-7B | 0.060 | +3.1 | −4.6 | 0.464 | 0.765 | **0.814** | 84 |
| SFT | 0.035 | +1.1 | −3.6 | 0.628 | 0.730 | **0.814** | 49 |
| SFT+DPO | 0.026 | +1.2 | −3.3 | 0.651 | 0.673 | **0.725** | 36 |
| SFT+DPO+RLVR (Instruct) | 0.035 | +0.9 | −2.7 | 0.652 | 0.644 | **0.726** | 49 |
| RL-Zero (from base) | 0.065 | +2.3 | −3.1 | 0.583 | 0.791 | **0.827** | 91 |
| AlphaMed-7B | 0.071 | +1.6 | −3.0 | 0.586 | 0.755 | **0.797** | 100 |
| HuatuoGPT-o1-7B | 0.093 | +2.6 | −5.7 | 0.653 | 0.708 | **0.756** | 130 |
| m1-7B | 0.084 | +2.9 | −7.9 | 0.650 | 0.747 | **0.800** | 117 |
| AlphaMed-7B (base-RL) | 0.054 | +2.5 | −4.2 | 0.672 | 0.733 | **0.789** | 75 |

Table 13: Aggregation and uncertainty on Phenopacket Store (1,401 real rare-disease cases, 32 samples, $T$=0.7). An answer counts as correct if it is the diagnosis or a closely related Mondo class. Columns as in Table 8; Correct answers: number of correct single-call answers, the positives for the AUROCs. Bold: best signal.

### 4.13 Stated and sampled readouts on real rare-disease cases

Phenopacket Store cases are real, but they have no reference differential and accuracy is near the floor, so we can test only the consequences of §4.7, and we count an answer as correct if it is the diagnosis or a closely related Mondo class (exact matches give only 8–31 correct single-call answers per model). Top answers are grouped by Mondo class for voting and agreement (Table 13). The DDXPlus shifts reappear, without a crossover. Majority voting over 32 samples helps the base models by 3.1–4.6 points but the final post-trained models by only 0.4–1.5 (Qwen3-8B +0.4, n.s.). Post-training weakens agreement as an error signal (Qwen3-8B 0.830 → 0.763, Qwen2.5-7B 0.845 → 0.745, OLMo 3 7B 0.814 → 0.726) and strengthens the stated probability (0.567 → 0.707, 0.598 → 0.717, 0.464 → 0.652). Agreement beats the stated probability by 0.25–0.35 AUROC for the base models and by only 0.03–0.08 for the post-trained ones, significantly only for Qwen3-8B (−0.056 $[-0.120, -0.002]$). On these open-ended cases the three most frequent of 32 sampled answers still contain a correct diagnosis more often than one list's top three, but the gap shrinks from 4.6–7.0 points for the base models to 2.7–3.6 for Qwen2.5-7B-Instruct and OLMo-3-7B-Instruct and to a tie for Qwen3-8B (+0.4 $[-0.8, +1.8]$). The medical models behave like their post-trained parent, with agreement the better signal for all of them.

## 5 Training without collapsing the samples

The stated list survives post-training, but sampled diversity still matters: self-consistency and other sampling-based methods depend on it (§4.7), and so does reinforcement learning itself, since GRPO learns nothing from a group whose samples agree. Can reinforcement learning raise diagnostic accuracy without collapsing the sampled differential? We fine-tune Qwen3-1.7B with GRPO on DDXPlus training patients (9,800 cases, 200 per condition; disjoint from the test split) and evaluate exactly as in §3.

**Setup.** We train LoRA adapters ($r$=32) with GRPO for 200 steps. Each step uses 16 prompts and 16 sampled completions per prompt, generated with vLLM on the same GPU. A first sweep with

a common aggressive configuration (learning rate $2\times10^{-5}$, no KL penalty, 8 samples per prompt, per-group reward standardization) was unstable. In three of seven runs, the policy collapsed to an input-independent answer: one condition for 88–100% of all patients, with completions shrinking to about 25 tokens. The starting policy is nearly deterministic (token entropy ≈0.06), so most GRPO groups have zero reward variance, and noisy updates drive the collapse. We therefore use learning rate $5\times10^{-6}$, KL coefficient 0.04, no reward standardization (as in Puri et al. [24]), and 16 samples per prompt. A monitor tracks the number of distinct top-1 answers per case within each batch. All runs in this configuration were stable.

**Rewards.** Every recipe includes a format reward: the answer must parse into a list of distinct conditions with valid probabilities. `acc` is standard RLVR: 1 if the top-1 answer is the true pathology. `dist` adds two per-sample distributional terms to `acc`: a Brier score of the stated distribution against the truth, and the overlap of the stated distribution with the reference differential. Both are computed from a *single* completion. They cannot see that the policy gives the same answer in every sample, which is exactly the collapse of §4.2. `gm` adds to `acc` a *group-level* reward over the GRPO group of $n$ samples for a case. If sample $i$'s top-1 answer is $a$, and $k_a$ samples in the group give $a$,

$$r_i^{\mathrm{gm}} = \mathrm{clip}\big(n\,p_{\mathrm{ref}}(a) - (k_a - 1),\ 0,\ 1\big), \tag{1}$$

where $p_{\mathrm{ref}}$ is the reference differential. This is sample $i$'s marginal contribution to $n\sum_c \min(q_c, p_{\mathrm{ref}}(c))$, the overlap between the group's empirical answer distribution $q$ and the reference. An answer earns credit while its condition is under-represented relative to its reference mass, and none once it is over-represented. In contrast to multi-answer rewards that score a set inside one completion [24], `gm` acts on the distribution of *sampled* answers, the quantity that collapses. Because the group reward fires only when a group contains dissenting samples, we also test a boosted variant, `gm`$^+$: `gm` weight 3 and training temperature 1.2. As baselines we adapt the multi-answer rewards of Puri et al. [24] to our prompt. Multi-RLVR scores the share of listed conditions that belong to the reference differential, treated as a set of acceptable answers; Multi-RLCR adds their set-level Brier term on the per-answer probabilities. Both zero the reward for duplicate or invalid answers, as in the original, and use the same format reward, optimizer and settings as our recipes.

**Results.** Table 15 reports means over two seeds; intervals bootstrap cases and do not include seed-to-seed variation. All our recipes raise top-1 accuracy by 8–10 points; the multi-answer baselines barely move it (below). The per-sample recipes (`acc`, `dist`) slightly *reduce* diversity relative to the starting model, the sharpening documented in §4.1. The group reward removes this. `gm` matches the starting model's diversity, and `gm`$^+$ exceeds it while also being the most accurate recipe. Against `acc`, with paired bootstraps over cases averaged over seeds, `gm`$^+$ improves:

- pass@1 by +1.6 [+1.0, +2.3] and pass@32 by +1.7 [+1.0, +2.6] points;
- distinct answers by +0.10 [+0.07, +0.13] and entropy by +0.034 [+0.022, +0.047];
- coverage@32 by +1.5 [+1.0, +2.1] points, and TV by −0.014 [−0.019, −0.010].

Plain `gm` improves on `acc` in the same direction with smaller margins (e.g. pass@1 +1.0 [+0.6, +1.6], entropy +0.013 [+0.005, +0.021]). Diversity increases monotonically with group-reward pressure, with no accuracy cost. The set-membership baselines of Puri et al. [24] train stably (their reward rises from 0.27–0.31 to 0.32–0.35) but barely change sampled behavior. pass@1 rises by under 1 point over the untrained model, and diversity is unchanged. Against them, `gm`$^+$ is better on every metric, by +8.9 [+7.4, +10.5] points at pass@1, +0.07 [+0.03, +0.11] distinct answers and +3.0

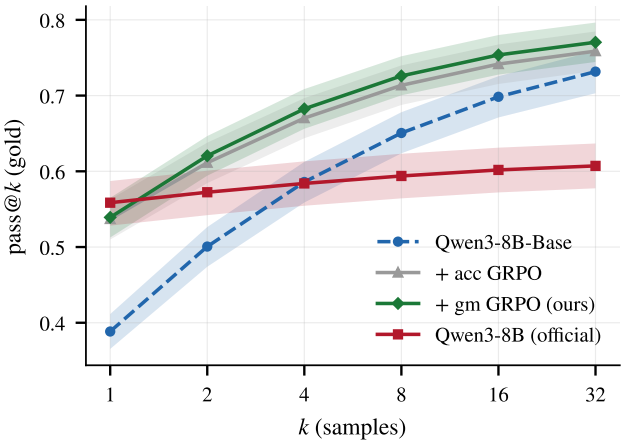



Figure 5: pass@$k$ on DDXPlus ($T$=0.7, 980 cases) from the same pretrained weights: Qwen3-8B-Base, our GRPO runs (pooled over two seeds), and the official post-trained Qwen3-8B, with 95% case-bootstrap bands. Our RL lifts the whole curve; official post-training trades the right end for the left.

$[+2.3, +3.8]$ points of coverage@32 (vs. MULTI-RLVR; MULTI-RLCR is similar). What they do improve is the differential written *inside* one completion: the reference mass covered by a single listed differential rises to 0.317 (MULTI-RLVR), against 0.300 for `acc` and 0.309 for `gm`$^+$. The two objectives are complementary: a set reward shapes the list a model writes, while the group reward shapes which answer it commits to across samples, which is what collapses under post-training. The absolute effects are modest because the starting policy is already collapsed: there is little exploration for any reward to act on. None of the recipes changes rare-disease accuracy or confidence on Phenopacket Store, where a 1.7B model is at the floor; the same holds at 8B (§4.12).

**Starting from a diverse base model.** The 1.7B experiment starts from an already collapsed instruct model. We therefore repeat the comparison from Qwen3-8B-Base, the diverse base model of §4.1, with the same recipe and `acc` vs. `gm`. We compare against Qwen's own post-trained Qwen3-8B (Table 14; means of two seeds). Both RL recipes reach nearly the official model's top-1 accuracy (0.537 and 0.539 vs. 0.558; `gm`: $-1.9$ $[-3.9, +0.1]$ points). They keep more than twice its diversity (2.7–2.9 vs. 1.27 distinct answers) and beat it by 15–16 points at pass@32 (`gm`: $+16.3$ $[+14.0, +18.8]$). In this setting, then, the collapse produced by general post-training is not required for its top-1 accuracy (our runs train on in-distribution DDXPlus data, so this is an existence result, not a like-for-like comparison of recipes). Both runs still narrow the base model somewhat, and the group reward narrows it less. At equal pass@1 ($+0.2$, n.s.), `gm` beats `acc` on distinct answers $+0.14$ $[+0.11, +0.17]$, entropy $+0.035$ $[+0.029, +0.041]$, coverage@32 $+1.1$ $[+0.8, +1.5]$ points, and pass@32 $+1.2$ $[+0.5, +1.9]$. Each seed reproduces the ordering. The severe tier tells the same story (Table 3): official post-training removes 9.8 points of can't-miss consideration, `acc` 2.8, and `gm` 1.5 (not significant). Figure 5 shows the pass@$k$ curves.

## 6 Prompts, vignettes and grading

**DDXPlus.** Each patient's coded evidence is rendered by a fixed template into a vignette of the patient's answers and relevant history; we fix known mistranslations in the English evidence descriptions. An example (test case with true pathology chronic rhinosinusitis; the reference differential

| Model | pass@1 | pass@32 | Distinct | Entropy | Cov@32 | TV↓ |
|---|---|---|---|---|---|---|
| Qwen3-8B-Base (start) | 0.389 | 0.732 | **3.77** | **0.78** | **0.323** | 0.749 |
| + `acc` GRPO (ours) | 0.537 | 0.759 | 2.72 | 0.52 | 0.297 | 0.750 |
| + `gm` GRPO (ours) | 0.539 | **0.770** | 2.86 | 0.56 | 0.308 | **0.742** |
| Qwen3-8B (official post-training) | **0.558** | 0.607 | 1.27 | 0.11 | 0.217 | 0.793 |

Table 14: GRPO from Qwen3-8B-Base vs. the official post-trained Qwen3-8B on DDXPlus (980 cases, 32 samples, $T$=0.7); GRPO rows are means of two seeds. Our RL nearly matches the official model's top-1 accuracy while keeping more than twice its diversity; the group reward preserves more than `acc`. The base row comes from this experiment's own sampling run, so it differs slightly from Table 2.

| Recipe | pass@1 | pass@32 | Distinct | Entropy | Cov@32 | TV↓ |
|---|---|---|---|---|---|---|
| Qwen3-1.7B (untrained) | 0.378 | 0.435 | 1.41 | 0.151 | 0.162 | 0.850 |
| `acc` | 0.458 | 0.499 | 1.36 | 0.142 | 0.181 | 0.829 |
| `dist` | 0.458 | 0.502 | 1.34 | 0.131 | 0.181 | 0.829 |
| MULTI-RLVR [24] | 0.385 | 0.443 | 1.39 | 0.153 | 0.166 | 0.845 |
| MULTI-RLCR [24] | 0.383 | 0.441 | 1.39 | 0.150 | 0.165 | 0.847 |
| `gm` | 0.469 | 0.510 | 1.40 | 0.155 | 0.186 | 0.825 |
| `gm`$^+$ | **0.474** | **0.516** | **1.46** | **0.176** | **0.196** | **0.815** |

Table 15: GRPO on Qwen3-1.7B, evaluated on DDXPlus test (980 cases, 32 samples, $T$=0.7); means of two seeds. `gm`$^+$: group-reward weight 3, training temperature 1.2. Per-sample rewards slightly narrow the model; set-membership rewards barely move it; the group reward preserves, then widens, its differential while improving accuracy.

gives viral pharyngitis 0.17, chronic rhinosinusitis 0.16, cluster headache 0.15, acute rhinosinusitis 0.14, bronchitis 0.14, URTI 0.12):

```
A 13-year-old female presents for evaluation. Symptoms (patient answers):
Do you have pain somewhere, related to your reason for consulting? Yes.
Characterize your pain: burning. Do you feel pain somewhere? mouth, forehead,
cheek(R), cheek(L), nose. How intense is the pain? (scale 0-10) 3. […] Have
you lost your sense of smell? Yes. Do you have nasal congestion or a clear
runny nose? Yes. Do you have a cough? Yes. Relevant history and risk factors:
Have you had a cold in the last 2 weeks? Yes. […] Do you have polyps in your
nose? Yes. […]
```

System prompt: `You are an experienced emergency and primary care physician.` The user turn is the vignette followed by: `Based on the patient's answers above, give your differential diagnosis. Choose ONLY from this list of possible conditions: [the 49 conditions, one per line]. List the 5 most likely conditions from the list, most likely first, with your probability that each is the correct diagnosis. Use the condition names exactly as written in the list. End your answer with a JSON array in exactly this format: [{"diagnosis": "<name>", "probability": <number between 0 and 1>}, ...]`. The full-distribution prompt (§4.8) instead asks for a probability on every condition, summing to one, most likely first. The plain and CoT variants are described in §4.3. Base models receive the same chat-formatted prompt; OLMo base models use their SFT checkpoint's chat template.

**Phenopacket Store.** The vignette lists the case's present and (up to ten) excluded HPO phenotypes with sex and age. System prompt: `You are an expert clinical geneticist and diagnostician.` Instructions: `List the 5 most likely specific diagnoses (use the specific disease or subtype name, e.g. 'Spastic paraplegia 7' rather than 'hereditary spastic paraplegia', when you can), most likely first, with your probability that each is the correct diagnosis`, followed by the same JSON format.

**Grading open-ended names.** Each predicted name is mapped to Mondo by exact lookup against Mondo labels and synonyms (also trying any parenthesized abbreviation and the name without it), then by fuzzy matching (token-sort ratio $\geq 92$). It is *exact* if it maps to the Mondo class of the case's OMIM diagnosis, and *related* if it maps to a specific ancestor group of it (at most two is-a steps up, with at most 150 descendants) or to one of its subtypes; siblings sharing only a broad parent are wrong. The grader passes hand-written checks (e.g. "Hereditary spastic paraplegia type 7 (SPG7)" is exact for SPG7, "Hereditary spastic paraplegia" is related, and Turner syndrome is wrong for Jacobsen syndrome), but we did not validate it against clinician grading, so errors at the scale of the 1–2 point effects we report on this dataset are possible.

## 7 Limitations

DDXPlus is synthetic: its reference differential is the posterior of a rule-based engine, and the simulated pathology is not always ranked first in it (it is for "more than 70%" of patients [7, §3.5]), so we treat TV and coverage as distance to a *reference* posterior rather than to clinical truth. Base models more often give invalid answers. We count these as errors, which makes the large-$k$ losses conservative but inflates post-trained gains at small $k$; conditioning on valid samples removes a quarter to a third of the $T$=1 pass@1 gain (§4.3). Our base/post-trained comparisons cover two open families at 7–32B; prompt robustness of the sampled readout is tested on one pair, and the collapse at large $k$ largely disappears when the model is asked to reason first (§4.3). The OLMo base models give invalid answers on 27–47% of samples, so their format-controlled comparisons are weaker than Qwen's. We evaluate one RL-Zero checkpoint, at 7B, trained with an LM-judge reward on chat prompts rather than a diagnosis reward, and the 32B models only at $T$=0.7. Phenopacket cases are drawn from published literature and may be in pretraining data, which we partly address with era splits. The stated readout renormalizes each list's probabilities over the conditions it names, and the single-call answer is a model's first valid sample. A full-distribution prompt gives the same picture (§4.8), but other elicitations may differ. Which readout is the better error signal is model-dependent; the audit of §4.10 makes it checkable with 100 labeled cases but does not remove the dependence, and we validate it only within DDXPlus. The test set is class-balanced (20 cases per condition), so accuracies do not reflect real prevalence, and the prompt lists all 49 conditions, so DDXPlus is a closed-set task. We do not evaluate clinical utility with clinicians.

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